

# PS1 - IO

Tate Mason

```
# Packages
library(tidyverse)

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
library(lubridate)
library(stargazer)
```

Please cite as:

Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.  
R package version 5.2.3. <https://CRAN.R-project.org/package=stargazer>

```
library(fixest)
library(here)
```

here() starts at /Users/tate/SchoolWork

```

library(readr)
library(tidyr)
library(stringr)

# Data
iri <- read_tsv("~/SchoolWork/Y2S1/I0/PS1/IRI.csv", show_col_types = FALSE)
names(iri) <- gsub("^t", "", names(iri)) # Clean names
iri <- iri %>%
  mutate(
    market_id = interaction(store_id, week_id, drop = TRUE),
    date_id = interaction(year, month, drop = TRUE),
    date = as.Date(paste(year, month, "01", sep = "-"))
  )

```

## Question 1:

### 1.1

```

# Distinct counts per market
brands_per_market <- iri %>%
  distinct(market_id, brand) %>%
  count(market_id, name = "n_brands")

mfrs_per_market <- iri %>%
  distinct(market_id, parent) %>%
  count(market_id, name = "n_mfrs")

# Total market sales per market
total_sales_market <- iri %>%
  group_by(market_id) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Brand-level sales (market x brand)
brand_sales_market <- iri %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Per-market price & characteristics (means within market)

```

```

market_chars <- iri %>%
  group_by(market_id) %>%
  summarise(
    price_mean      = mean(price, na.rm = TRUE),
    fiber_mean      = mean(fiber, na.rm = TRUE),
    sugars_mean     = mean(sugars, na.rm = TRUE),
    flavored_mean   = mean(as.numeric(flavored), na.rm = TRUE),
    fortified_mean  = mean(as.numeric(fortified), na.rm = TRUE),
    .groups = "drop"
  )

# Helper to summarise numeric vectors
summarise_vec <- function(x) {
  tibble(
    mean  = mean(x, na.rm = TRUE),
    sd    = sd(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    max   = max(x, na.rm = TRUE)
  )
}

# Tables for 1.1
tbl_counts <- bind_cols(
  variable = c("n_brands", "n_mfrs"),
  bind_rows(
    summarise_vec(brands_per_market$n_brands),
    summarise_vec(mfrs_per_market$n_mfrs)
  )
)

tbl_total_sales <- bind_cols(
  variable = "total_sales (per market)",
  summarise_vec(total_sales_market$total_sales)
)

tbl_brand_sales <- bind_cols(
  variable = "brand_sales (per market-brand)",
  summarise_vec(brand_sales_market$brand_sales)
)

tbl_price_chars <- tribble(
  ~variable,          ~mean, ~sd, ~median, ~max,

```

```

"price (market mean)",
  summarise_vec(market_chars$price_mean)$mean,
  summarise_vec(market_chars$price_mean)$sd,
  summarise_vec(market_chars$price_mean)$median,
  summarise_vec(market_chars$price_mean)$max,
"fiber (market mean)",
  summarise_vec(market_chars$fiber_mean)$mean,
  summarise_vec(market_chars$fiber_mean)$sd,
  summarise_vec(market_chars$fiber_mean)$median,
  summarise_vec(market_chars$fiber_mean)$max,
"sugars (market mean)",
  summarise_vec(market_chars$sugars_mean)$mean,
  summarise_vec(market_chars$sugars_mean)$sd,
  summarise_vec(market_chars$sugars_mean)$median,
  summarise_vec(market_chars$sugars_mean)$max,
"flavored share (market mean)",
  summarise_vec(market_chars$flavored_mean)$mean,
  summarise_vec(market_chars$flavored_mean)$sd,
  summarise_vec(market_chars$flavored_mean)$median,
  summarise_vec(market_chars$flavored_mean)$max,
"fortified share (market mean)",
  summarise_vec(market_chars$fortified_mean)$mean,
  summarise_vec(market_chars$fortified_mean)$sd,
  summarise_vec(market_chars$fortified_mean)$median,
  summarise_vec(market_chars$fortified_mean)$max
)

summary_11 <- bind_rows(
  tbl_counts,
  tbl_total_sales,
  tbl_brand_sales,
  tbl_price_chars
)

summary_11

```

```

# A tibble: 9 x 5
  variable          mean      sd  median    max
  <chr>          <dbl>    <dbl>  <dbl>  <dbl>
1 n_brands      34.9      4.82    37     39
2 n_mfrs        3.95     0.259    4      4
3 total_sales (per market) 15555.  12358.  12557.  84567.

```

|                                  |       |        |       |        |
|----------------------------------|-------|--------|-------|--------|
| 4 brand_sales (per market-brand) | 446.  | 726.   | 238.  | 26470. |
| 5 price (market mean)            | 0.235 | 0.0265 | 0.233 | 0.325  |
| 6 fiber (market mean)            | 1.30  | 0.0874 | 1.31  | 1.67   |
| 7 sugars (market mean)           | 8.31  | 0.205  | 8.33  | 10.2   |
| 8 flavored share (market mean)   | 0.640 | 0.0265 | 0.641 | 0.857  |
| 9 fortified share (market mean)  | 0.485 | 0.0341 | 0.487 | 0.667  |

```
# If you want a file:
# write_csv(summary_11, "summary_11_descriptives.csv")
```

## 1.2

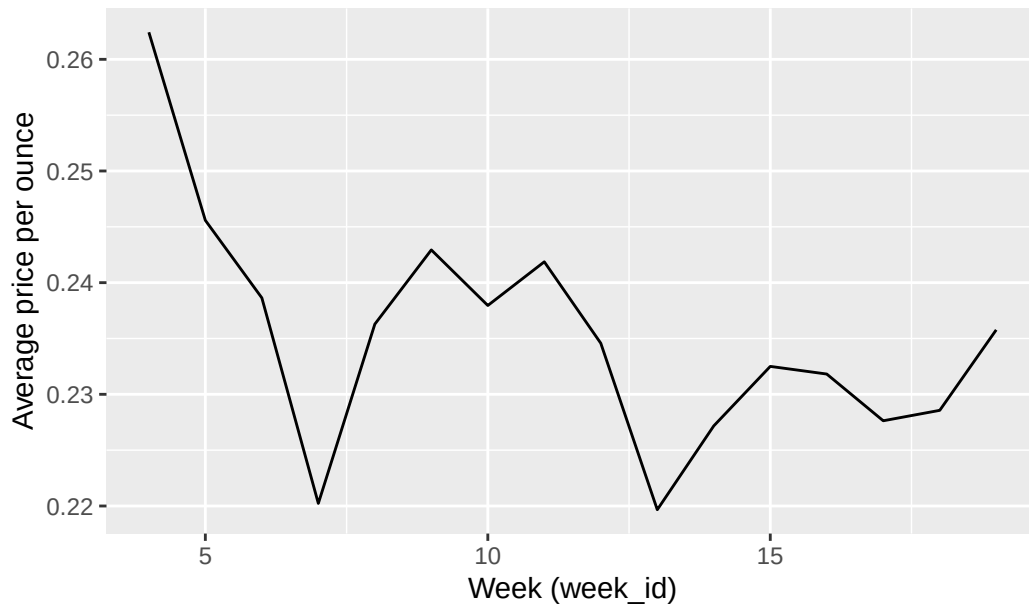
```
ts_simple <- iri %>%
  group_by(week_id) %>%
  summarise(
    avg_price = mean(price, na.rm = TRUE),
    avg_qty   = mean(quantity, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(week_id)

p_price <- ggplot(ts_simple, aes(x = week_id, y = avg_price)) +
  geom_line() +
  labs(title = "Average Cereal Price over Time",
       x = "Week (week_id)", y = "Average price per ounce")

p_qty <- ggplot(ts_simple, aes(x = week_id, y = avg_qty)) +
  geom_line() +
  labs(title = "Average Cereal Sales (Quantity) over Time",
       x = "Week (week_id)", y = "Average quantity (lbs)")

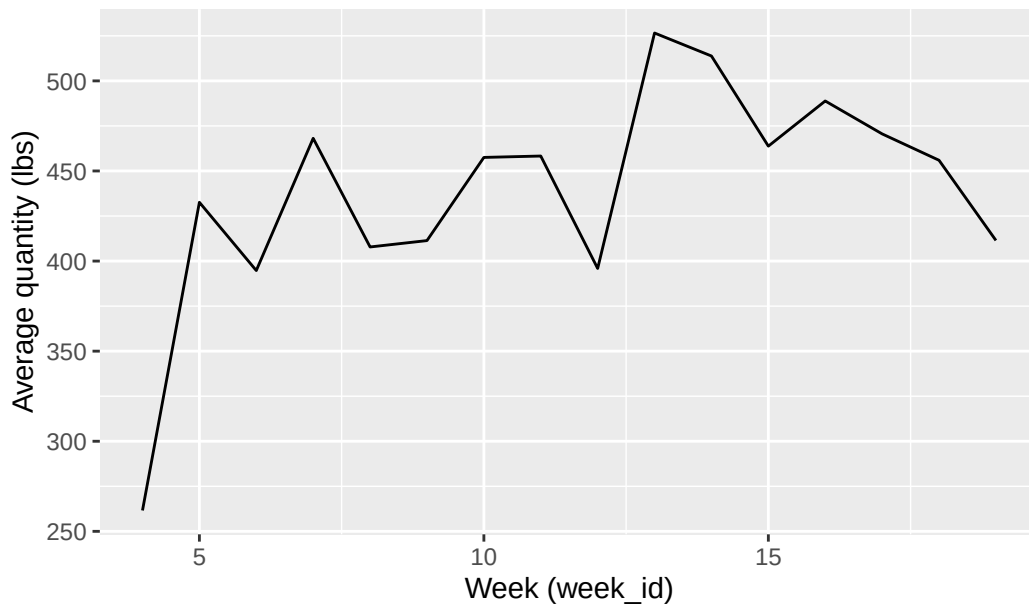
p_price
```

Average Cereal Price over Time



p\_qty

Average Cereal Sales (Quantity) over Time



```
# Save figures
ggsave("q12-ts-1.pdf", p_price, width = 10, height = 6, dpi = 300)
```

```

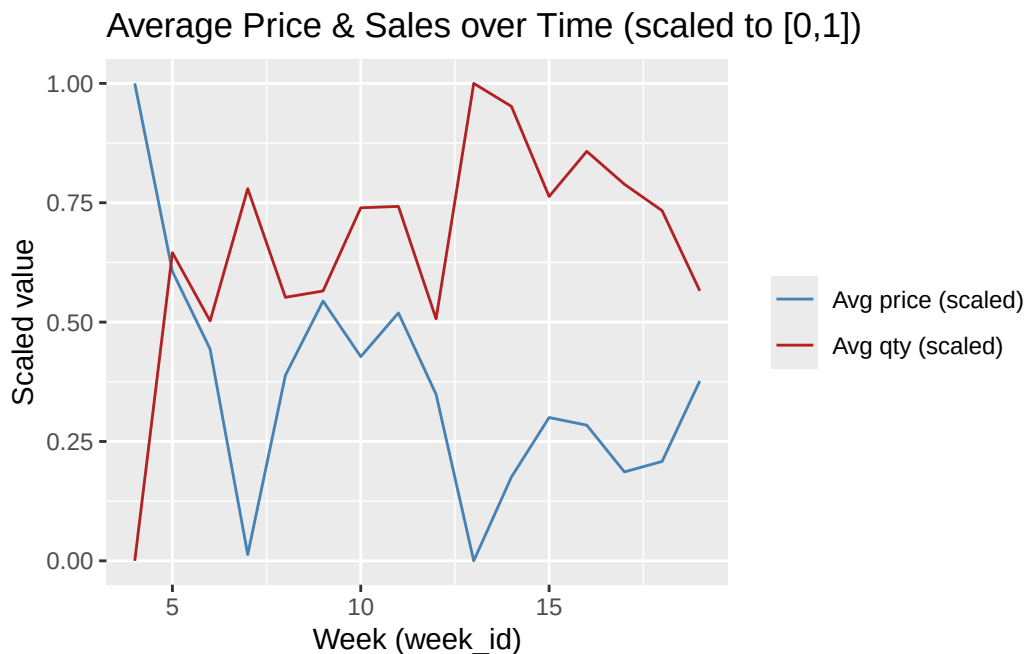
ggsave("q12-ts-2.pdf", p_qty, width = 10, height = 6, dpi = 300)

# Dual-axis style (scaled)
ts_long_scaled <- ts_simple %>%
  mutate(
    price_scaled = (avg_price - min(avg_price, na.rm = TRUE)) /
      (max(avg_price, na.rm = TRUE) - min(avg_price, na.rm = TRUE)),
    qty_scaled    = (avg_qty - min(avg_qty, na.rm = TRUE)) /
      (max(avg_qty, na.rm = TRUE) - min(avg_qty, na.rm = TRUE))
  ) %>%
  select(week_id, price_scaled, qty_scaled) %>%
  pivot_longer(-week_id, names_to = "series", values_to = "value")

p_dual <- ggplot(ts_long_scaled, aes(x = week_id, y = value, color = series)) +
  geom_line() +
  scale_color_manual(values = c("price_scaled" = "steelblue", "qty_scaled" = "firebrick"),
    labels = c("Avg price (scaled)", "Avg qty (scaled)")) +
  labs(title = "Average Price & Sales over Time (scaled to [0,1])",
    x = "Week (week_id)", y = "Scaled value", color = NULL)

p_dual

```



```
ggsave("q12-ts-3.png", p_dual, width = 10, height = 6, dpi = 300)
```

### 1.3

```
# Brand HHI per market
brand_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_brand = if_else(total_mkt_sales > 0, brand_sales / total_mkt_sales, 0))

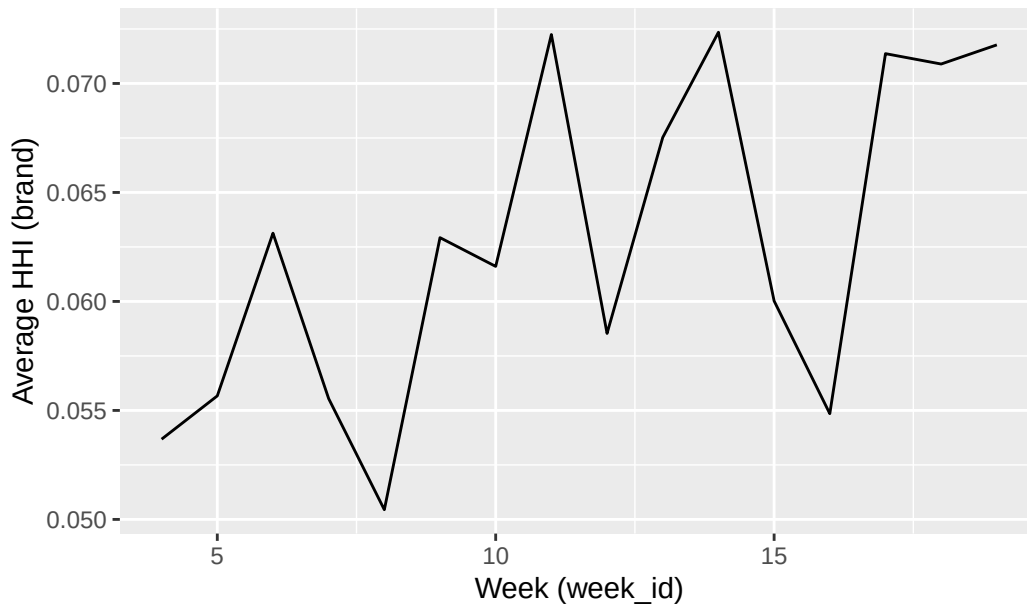
hhi_brand_market <- brand_shares %>%
  group_by(market_id) %>%
  summarise(HHI_brand = sum(share_brand^2, na.rm = TRUE), .groups = "drop")

hhi_brand_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_brand_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_brand = mean(HHI_brand, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)

p_hhi_brand <- ggplot(hhi_brand_week, aes(x = week_id, y = avg_HHI_brand)) +
  geom_line() +
  labs(title = "Average Brand-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (brand)")

p_hhi_brand
```

Average Brand-Level HHI Over Time



```
ggsave("q13-hhi-1.pdf", p_hhi_brand, width = 10, height = 6, dpi = 300)

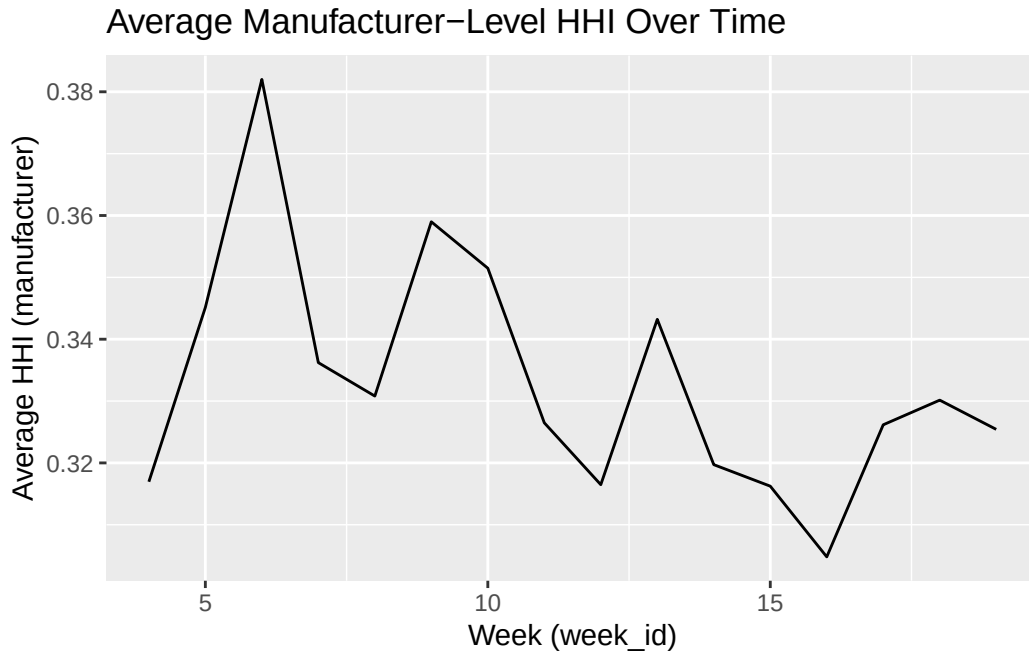
# Manufacturer (parent) HHI per market
parent_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, parent) %>%
  summarise(parent_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_parent = if_else(total_mkt_sales > 0, parent_sales / total_mkt_sales, 0))

hhi_parent_market <- parent_shares %>%
  group_by(market_id) %>%
  summarise(HHI_parent = sum(share_parent^2, na.rm = TRUE), .groups = "drop")

hhi_parent_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_parent_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_parent = mean(HHI_parent, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)
```

```
p_hhi_parent <- ggplot(hhi_parent_week, aes(x = week_id, y = avg_HHI_parent)) +
  geom_line() +
  labs(title = "Average Manufacturer-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (manufacturer)")

p_hhi_parent
```



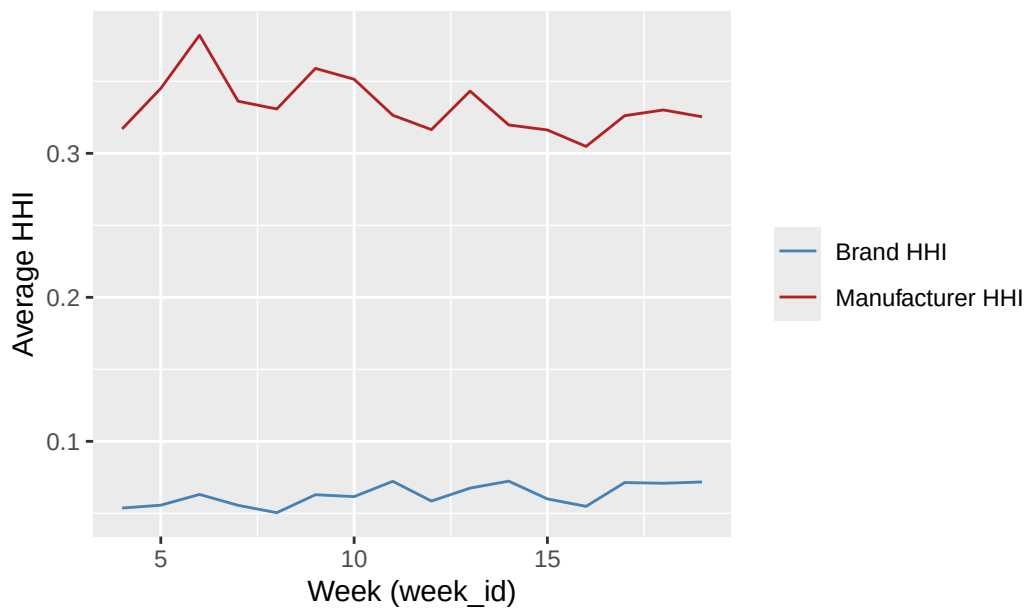
```
ggsave("q13-hhi-2.pdf", p_hhi_parent, width = 10, height = 6, dpi = 300)

# Combined comparison
hhi_both <- hhi_brand_week %>% inner_join(hhi_parent_week, by = "week_id")

p_hhi_both <- ggplot(hhi_both, aes(x = week_id)) +
  geom_line(aes(y = avg_HHI_brand, color = "Brand HHI")) +
  geom_line(aes(y = avg_HHI_parent, color = "Manufacturer HHI")) +
  scale_color_manual(values = c("Brand HHI" = "steelblue", "Manufacturer HHI" = "firebrick")) +
  labs(title = "Average HHI Over Time (Brand vs Manufacturer)",
       x = "Week (week_id)", y = "Average HHI", color = NULL)

p_hhi_both
```

Average HHI Over Time (Brand vs Manufacturer)



```
ggsave("q13-hhi-3.pdf", p_hhi_both, width = 10, height = 6, dpi = 300)
```

## Question 2:

### 2.1

```
# Prepare data for regression
reg_data <- iri %>%
  group_by(market_id) %>%
  mutate(
    market_fe = as.factor(store_id),
    manufacturer_fe = as.factor(parent),
    time_fe = as.factor(week_id),
    brand_fe = as.factor(brand),
    p_jmt = price,
    total_sales = sum(quantity, na.rm = TRUE),
    sj = quantity / M,
    s0 = 1 - (total_sales / M),
    u = log(sj) - log(s0)
  ) %>%
  ungroup()
```

Our estimation model is thus:

$$u_{jmt} = \alpha p_{jmt} + \beta \mathbb{X}_{jmt} + \xi_{jmt}$$

where  $\mathbb{X}_{jmt}$  includes the product characteristics fiber, sugars, flavored, and fortified, as well as fixed effects for market, manufacturer, and time. and  $u_{jmt}$  is the log-odds of product j's market share in market m at time t.

## 2.2

```
# Regression model
list <- c("fiber", "sugars", "flavored", "fortified",
         "market_fe", "manufacturer_fe", "time_fe")
reg_model <- feols(
  u ~ p_jmt + fiber + sugars + flavored + fortified |
  market_fe + manufacturer_fe + time_fe,
  data = reg_data
)

summary(reg_model)
```

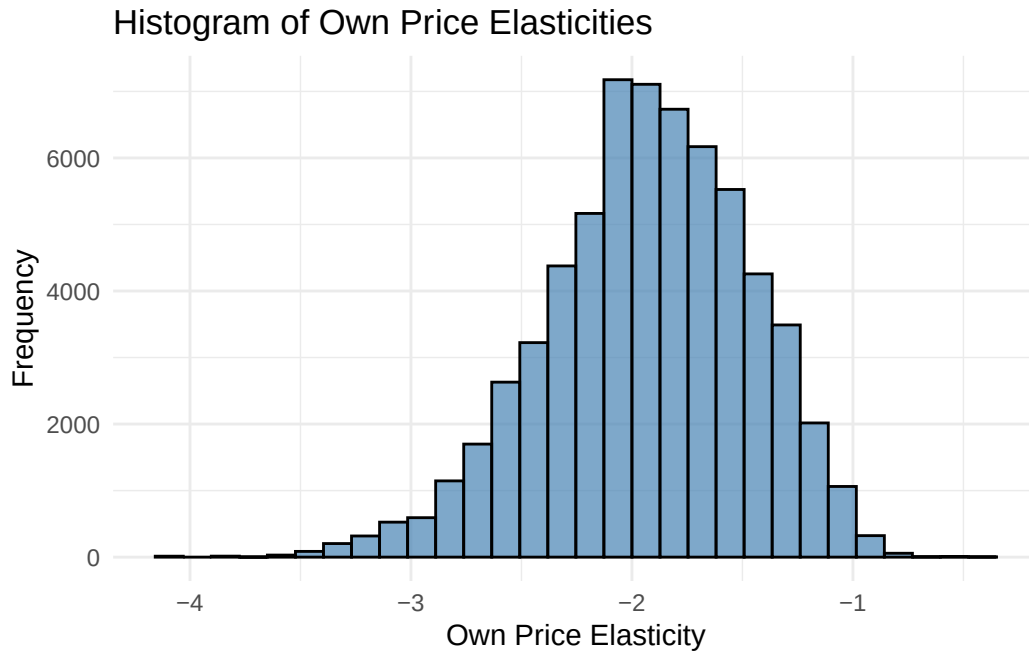
```
OLS estimation, Dep. Var.: u
Observations: 63,966
Fixed-effects: market_fe: 642, manufacturer_fe: 4, time_fe: 16
Standard-errors: IID
```

|           | Estimate  | Std. Error | t value   | Pr(> t )      |
|-----------|-----------|------------|-----------|---------------|
| p_jmt     | -8.314431 | 0.083677   | -99.36340 | < 2.2e-16 *** |
| fiber     | -0.118828 | 0.003300   | -36.01266 | < 2.2e-16 *** |
| sugars    | -0.043445 | 0.001077   | -40.33986 | < 2.2e-16 *** |
| flavored  | 0.192915  | 0.008305   | 23.22877  | < 2.2e-16 *** |
| fortified | 0.004322  | 0.009132   | 0.47327   | 0.63602       |

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.899505      Adj. R2: 0.248711
                Within R2: 0.151216
```

```
# Histogram of own price elasticities
reg_data <- reg_data %>%
  mutate(own_price_elasticity = coef(reg_model)["p_jmt"] * p_jmt * (1 - sj))
```

```
ggplot(reg_data, aes(x = own_price_elasticity)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities",
       x = "Own Price Elasticity", y = "Frequency") +
  theme_minimal()
```



```
ggsave("q2-2-1.pdf", width = 10, height = 6, dpi = 300)
```

## 2.3

```
# 2SLS with Price Instrument: Price*Quantity of Sugar

reg_data <- reg_data %>%
  mutate(
    z = sugars * sugar_price,
    z = log(z)
  )
stage2 <- feols(u ~ fiber + sugars + flavored + fortified |
               market_fe + manufacturer_fe + time_fe | p_jmt ~ z,
```

```
data = reg_data)
summary(stage2)
```

```
TSLS estimation - Dep. Var.: u
                Endo.      : p_jmt
                Instr.     : z
```

```
Second stage: Dep. Var.: u
```

```
Observations: 63,966
```

```
Fixed-effects: market_fe: 642, manufacturer_fe: 4, time_fe: 16
```

```
Standard-errors: IID
```

|           | Estimate   | Std. Error | t value   | Pr(> t )   |     |
|-----------|------------|------------|-----------|------------|-----|
| fit_p_jmt | -62.766445 | 7.898427   | -7.94670  | 1.9468e-15 | *** |
| fiber     | -0.570755  | 0.066161   | -8.62678  | < 2.2e-16  | *** |
| sugars    | -0.120572  | 0.011575   | -10.41695 | < 2.2e-16  | *** |
| flavored  | 0.179014   | 0.023118   | 7.74347   | 9.8166e-15 | *** |
| fortified | 0.377270   | 0.059710   | 6.31838   | 2.6607e-10 | *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
RMSE: 2.49435      Adj. R2: 0.137296
```

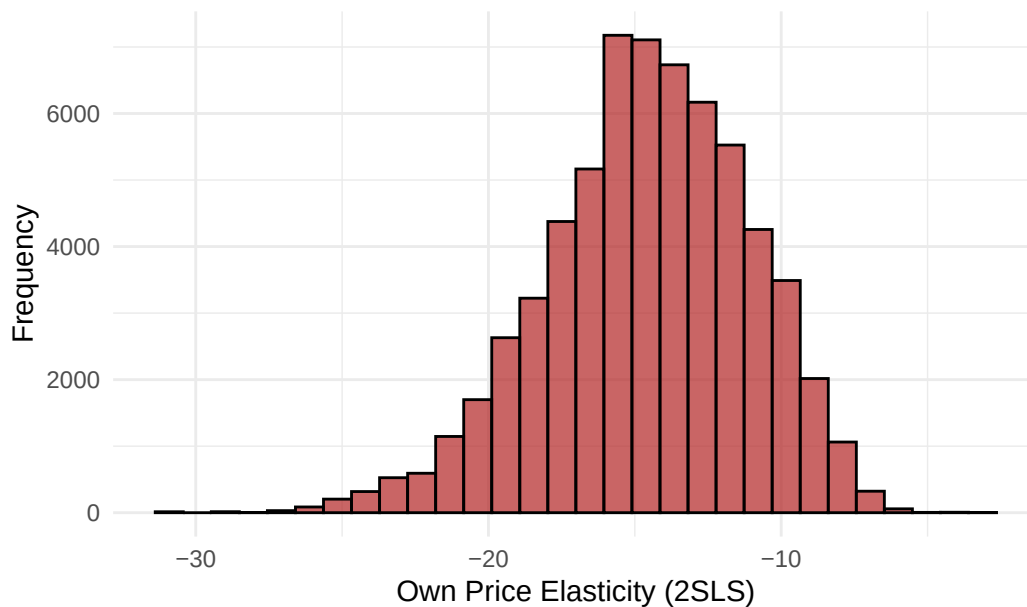
```
Within R2: 0.025342
```

```
F-test (1st stage), p_jmt: stat = 54.680, p = 1.436e-13, on 1 and 63,301 DoF.
```

```
Wu-Hausman: stat = 367.908, p < 2.2e-16 , on 1 and 63,300 DoF.
```

```
# Histogram of own price elasticities (2SLS)
reg_data <- reg_data %>%
  mutate(own_price_elasticity_2sls = coef(stage2)["fit_p_jmt"] * p_jmt * (1 - sj))
ggplot(reg_data, aes(x = own_price_elasticity_2sls)) +
  geom_histogram(bins = 30, fill = "firebrick", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities (2SLS)",
       x = "Own Price Elasticity (2SLS)", y = "Frequency") +
  theme_minimal()
```

Histogram of Own Price Elasticities (2SLS)



```
ggsave("q2-3-1.pdf", width = 10, height = 6, dpi = 300)
```

Using 2SLS results in estimates which are more negative than the OLS estimates, indicating that the OLS estimates were biased towards zero.

## 2.4

```
# Elasticity Matrix for Top 5 Brands by Sales
## Create average matrix by taking average of market level elasticity matrices

top_brands <- reg_data %>%
  group_by(brand) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(total_sales)) %>%
  slice_head(n = 5) %>%
  pull(brand)
elasticity_matrices <- list()
for (mkt in unique(reg_data$market_id)) {
  mkt_data <- reg_data %>% filter(market_id == mkt, brand %in% top_brands)
  n <- nrow(mkt_data)
  if (n < 5) next # Skip markets with fewer than 5 top brands
```

```

elas_matrix <- matrix(0, nrow = n, ncol = n)
rownames(elas_matrix) <- mkt_data$brand
colnames(elas_matrix) <- mkt_data$brand
for (i in 1:n) {
  for (j in 1:n) {
    if (i == j) {
      elas_matrix[i, j] <- coef(stage2)["fit_p_jmt"] * mkt_data$p_jmt[i] * (1 - mkt_data$sj[i])
    } else {
      elas_matrix[i, j] <- -coef(stage2)["fit_p_jmt"] * mkt_data$p_jmt[j] * mkt_data$sj[i]
    }
  }
}
elasticity_matrices[[as.character(mkt)]] <- elas_matrix
}
# Average elasticity matrix
avg_elasticity_matrix <- Reduce("+", elasticity_matrices) / length(elasticity_matrices)
avg_elasticity_matrix

```

|                               |                               |             |
|-------------------------------|-------------------------------|-------------|
|                               | KELLOGGS FROSTED MINI WHEATS  |             |
| KELLOGGS FROSTED MINI WHEATS  |                               | -13.1423398 |
| GENERAL MILLS CHEERIOS        |                               | 0.2615167   |
| GENERAL MILLS HONEY NUT CHEER |                               | 0.2654592   |
| POST HONEY BUNCHES OF OATS    |                               | 0.2657008   |
| KELLOGGS FROSTED FLAKES       |                               | 0.2576447   |
|                               | GENERAL MILLS CHEERIOS        |             |
| KELLOGGS FROSTED MINI WHEATS  | 0.2638177                     |             |
| GENERAL MILLS CHEERIOS        | -13.2669948                   |             |
| GENERAL MILLS HONEY NUT CHEER | 0.2671759                     |             |
| POST HONEY BUNCHES OF OATS    | 0.2681528                     |             |
| KELLOGGS FROSTED FLAKES       | 0.2604532                     |             |
|                               | GENERAL MILLS HONEY NUT CHEER |             |
| KELLOGGS FROSTED MINI WHEATS  | 0.2650948                     |             |
| GENERAL MILLS CHEERIOS        | 0.2639448                     |             |
| GENERAL MILLS HONEY NUT CHEER | -13.2314072                   |             |
| POST HONEY BUNCHES OF OATS    | 0.2691572                     |             |
| KELLOGGS FROSTED FLAKES       | 0.2576187                     |             |
|                               | POST HONEY BUNCHES OF OATS    |             |
| KELLOGGS FROSTED MINI WHEATS  | 0.2623082                     |             |
| GENERAL MILLS CHEERIOS        | 0.2636535                     |             |
| GENERAL MILLS HONEY NUT CHEER | 0.2664261                     |             |
| POST HONEY BUNCHES OF OATS    | -13.2162577                   |             |
| KELLOGGS FROSTED FLAKES       | 0.2599669                     |             |

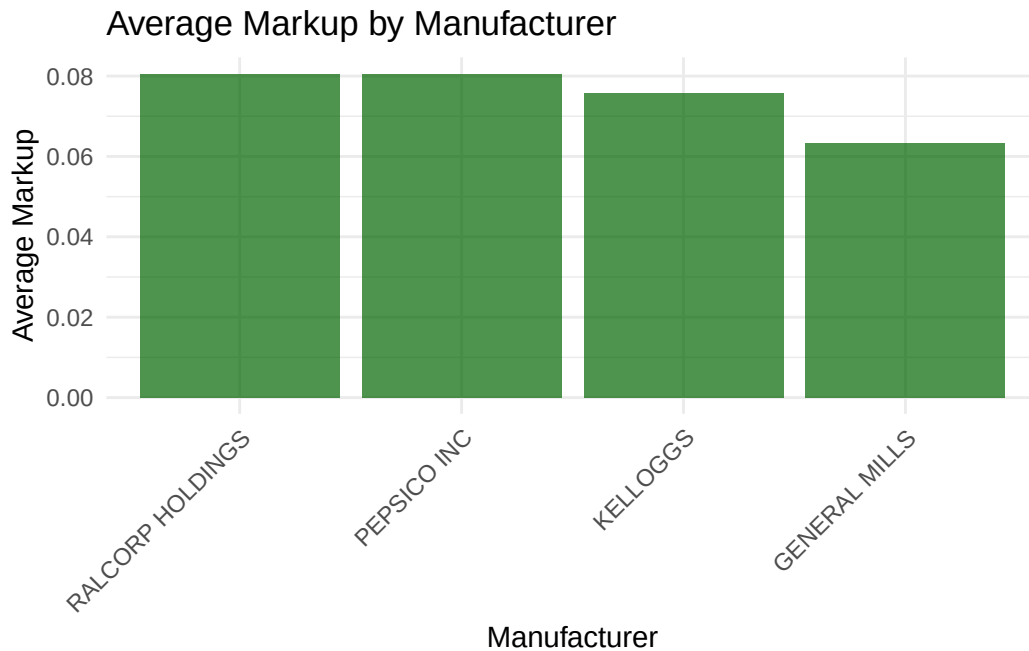
|                               | KELLOGGS FROSTED FLAKES |
|-------------------------------|-------------------------|
| KELLOGGS FROSTED MINI WHEATS  | 0.2669329               |
| GENERAL MILLS CHEERIOS        | 0.2632467               |
| GENERAL MILLS HONEY NUT CHEER | 0.2654629               |
| POST HONEY BUNCHES OF OATS    | 0.2690599               |
| KELLOGGS FROSTED FLAKES       | -13.2852126             |

## 2.5

```
# Calculaete markups for each manufacturer
reg_data <- reg_data %>%
  mutate(
    own_price_elasticity_2sls = coef(stage2)["fit_p_jmt"] * p_jmt * (1 - sj),
    markup = -1 / own_price_elasticity_2sls
  )
markup_summary <- reg_data %>%
  group_by(parent) %>%
  summarise(
    avg_markup = mean(markup, na.rm = TRUE),
    sd_markup = sd(markup, na.rm = TRUE),
    median_markup = median(markup, na.rm = TRUE),
    max_markup = max(markup, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(avg_markup))
markup_summary
```

```
# A tibble: 4 x 5
  parent      avg_markup sd_markup median_markup max_markup
  <chr>      <dbl>      <dbl>      <dbl>      <dbl>
1 RALCORP HOLDINGS 0.0806    0.0225    0.0764    0.320
2 PEPSICO INC      0.0805    0.0179    0.0786    0.226
3 KELLOGGS         0.0757    0.0182    0.0727    0.182
4 GENERAL MILLS    0.0634    0.0127    0.0627    0.269
```

```
ggplot(markup_summary, aes(x = reorder(parent, -avg_markup), y = avg_markup)) +
  geom_bar(stat = "identity", fill = "darkgreen", alpha = 0.7) +
  labs(title = "Average Markup by Manufacturer",
       x = "Manufacturer", y = "Average Markup") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
ggsave("q2-5-1.pdf", width = 10, height = 6, dpi = 300)
```

## 2.6

```
# Nested by segment - construct data and calculate conditional shares
reg_data <- reg_data %>%
  mutate(
    nest = as.factor(segment)
  ) %>%
  group_by(market_id, nest) %>%
  mutate(
    total_nest_sales = sum(quantity, na.rm = TRUE),
    share_con = if_else(total_nest_sales > 0, quantity / total_nest_sales, 0)
  ) %>%
  ungroup()
```

The estimating equation is now:

$$u_{jmt} = \alpha p_{jmt} + \beta \mathbb{X}_{jmt} + \sigma_g s_{j|g} + \xi_{jmt}$$

where  $s_{j|g}$  is the conditional share of product  $j$  within its nest  $g$ .

## 2.7

```
#2SLS with Price Instrument: Price*Quantity of Sugar and Conditional Share Instrument total p

reg_data <- reg_data %>%
  group_by(market_id) %>%
  mutate(
    z_cond = log(n_distinct(brand)),
    sjg = log(share_con),
    z = sugars * sugar_price,
    z = log(z)
  )
s2_nest <- feols(u ~ fiber + sugars + flavored + fortified |
  market_fe + manufacturer_fe + time_fe | p_jmt + sjg ~ z + z_cond,
  data = reg_data)
summary(s2_nest)
```

TSLS estimation - Dep. Var.: u

Endo. : p\_jmt, sjg

Instr. : z, z\_cond

Second stage: Dep. Var.: u

Observations: 63,966

Fixed-effects: market\_fe: 642, manufacturer\_fe: 4, time\_fe: 16

Standard-errors: IID

|           | Estimate   | Std. Error | t value  | Pr(> t )       |
|-----------|------------|------------|----------|----------------|
| fit_p_jmt | -54.396707 | 6.652299   | -8.17713 | 2.9602e-16 *** |
| fit_sjg   | 0.129922   | 0.078950   | 1.64563  | 9.9845e-02 .   |
| fiber     | -0.486850  | 0.059327   | -8.20618 | 2.3258e-16 *** |
| sugars    | -0.100596  | 0.012322   | -8.16377 | 3.3063e-16 *** |
| flavored  | 0.174628   | 0.020285   | 8.60859  | < 2.2e-16 ***  |
| fortified | 0.300395   | 0.055403   | 5.42196  | 5.9163e-08 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

RMSE: 2.15952 Adj. R2: 0.137644

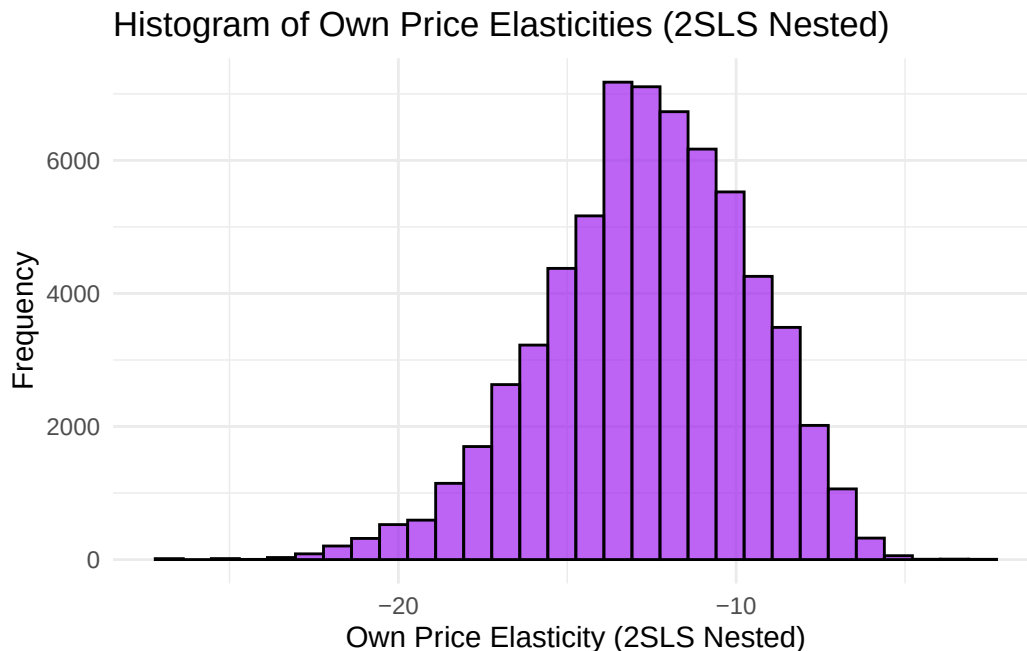
Within R2: 0.025751

F-test (1st stage), p\_jmt: stat = 32.803, p = 5.77e-15, on 2 and 63,300 DoF.

F-test (1st stage), sjg : stat = 472.922, p < 2.2e-16 , on 2 and 63,300 DoF.

Wu-Hausman: stat = 942.470, p < 2.2e-16 , on 2 and 63,298 DoF.

```
# Histogram of own price elasticities (2SLS Nested)
reg_data <- reg_data %>%
  mutate(own_price_elasticity_2sls_nest = coef(s2_nest)["fit_p_jmt"] * p_jmt * (1 - sj),
         sgj = log(coef(s2_nest)["fit_share_con"]))
ggplot(reg_data, aes(x = own_price_elasticity_2sls_nest)) +
  geom_histogram(bins = 30, fill = "purple", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities (2SLS Nested)",
       x = "Own Price Elasticity (2SLS Nested)", y = "Frequency") +
  theme_minimal()
```



```
ggsave("q2-7-1.pdf", width = 10, height = 6, dpi = 300)
```

## 2.8

```
# Elasticity Matrix for Top 5 Brands by Sales (Nested)
top_nest_brands <- reg_data %>%
  group_by(brand) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(total_sales)) %>%
  slice_head(n = 5) %>%
  pull(brand)
```

```

elasticity_nest_matrices <- list()
for (mkt in unique(reg_data$market_id)) {
  mkt_data <- reg_data %>% filter(market_id == mkt, brand %in% top_nest_brands)
  n <- nrow(mkt_data)
  if (n < 5) next # Skip markets with fewer than 5 top brands
  elas_matrix_nest <- matrix(0, nrow = n, ncol = n)
  rownames(elas_matrix_nest) <- mkt_data$brand
  colnames(elas_matrix_nest) <- mkt_data$brand
  for (i in 1:n) {
    for (j in 1:n) {
      if (i == j) {
        elas_matrix_nest[i, j] <- coef(s2_nest)["fit_p_jmt"] *
          mkt_data$p_jmt[i] * (1 / (1 - coef(s2_nest)["fit_sjg"])) *
          ((1 - coef(s2_nest)["fit_sjg"] * mkt_data$sjg[i]) - (1
            - coef(s2_nest)["fit_sjg"]) * mkt_data$sj[i]))
      } else if (mkt_data$nest[i] == mkt_data$nest[j]) {
        elas_matrix_nest[i, j] <- -coef(s2_nest)["fit_p_jmt"] *
          mkt_data$p_jmt[i] * (1 / (1 - coef(s2_nest)["fit_sjg"])) *
          ((1 - coef(s2_nest)["fit_sjg"] * mkt_data$sjg[i]) - (1
            - coef(s2_nest)["fit_sjg"]) * mkt_data$sj[i]))
      } else {
        elas_matrix_nest[i, j] <- coef(s2_nest)["fit_p_jmt"] *
          mkt_data$p_jmt[i] * (1 / (1 - coef(s2_nest)["fit_sjg"])) *
          ((1 - coef(s2_nest)["fit_sjg"] * mkt_data$sjg[i]) - (1 -
            coef(s2_nest)["fit_sjg"]) * mkt_data$sj[i]))
      }
    }
  }
  elasticity_nest_matrices[[as.character(mkt)]] <- elas_matrix_nest
}
# Average elasticity matrix (Nested)
avg_elasticity_nest_matrix <- Reduce("+", elasticity_nest_matrices) /
  length(elasticity_nest_matrices)
avg_elasticity_nest_matrix

```

|                               | KELLOGGS FROSTED MINI WHEATS |
|-------------------------------|------------------------------|
| KELLOGGS FROSTED MINI WHEATS  | -16.726258                   |
| GENERAL MILLS CHEERIOS        | 3.934947                     |
| GENERAL MILLS HONEY NUT CHEER | 3.884717                     |
| POST HONEY BUNCHES OF OATS    | 4.220221                     |
| KELLOGGS FROSTED FLAKES       | 3.412595                     |

|                               |                               |            |
|-------------------------------|-------------------------------|------------|
|                               | GENERAL MILLS CHEERIOS        |            |
| KELLOGGS FROSTED MINI WHEATS  |                               | 3.804315   |
| GENERAL MILLS CHEERIOS        |                               | -16.887474 |
| GENERAL MILLS HONEY NUT CHEER |                               | 4.515091   |
| POST HONEY BUNCHES OF OATS    |                               | 4.865219   |
| KELLOGGS FROSTED FLAKES       |                               | 4.027965   |
|                               | GENERAL MILLS HONEY NUT CHEER |            |
| KELLOGGS FROSTED MINI WHEATS  |                               | 3.769583   |
| GENERAL MILLS CHEERIOS        |                               | 4.375811   |
| GENERAL MILLS HONEY NUT CHEER |                               | -16.825855 |
| POST HONEY BUNCHES OF OATS    |                               | 4.791045   |
| KELLOGGS FROSTED FLAKES       |                               | 3.976238   |
|                               | POST HONEY BUNCHES OF OATS    |            |
| KELLOGGS FROSTED MINI WHEATS  |                               | 4.271827   |
| GENERAL MILLS CHEERIOS        |                               | 5.006285   |
| GENERAL MILLS HONEY NUT CHEER |                               | 4.941390   |
| POST HONEY BUNCHES OF OATS    |                               | -16.782152 |
| KELLOGGS FROSTED FLAKES       |                               | 4.387227   |
|                               | KELLOGGS FROSTED FLAKES       |            |
| KELLOGGS FROSTED MINI WHEATS  |                               | 3.286264   |
| GENERAL MILLS CHEERIOS        |                               | 3.794710   |
| GENERAL MILLS HONEY NUT CHEER |                               | 3.807976   |
| POST HONEY BUNCHES OF OATS    |                               | 4.280453   |
| KELLOGGS FROSTED FLAKES       |                               | -16.886185 |

## 2.9

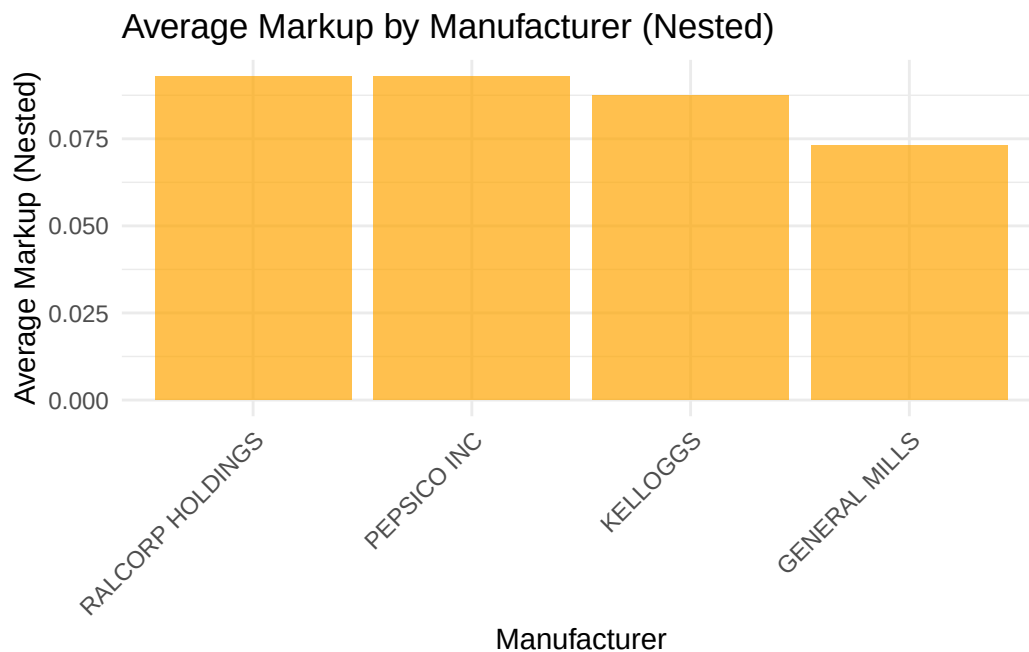
```
# Calculate average markups for each manufacturer (Nested)
reg_data <- reg_data %>%
  mutate(
    ope_nest = coef(s2_nest)["fit_p_jmt"] * p_jmt * (1 - sj),
    markup_nest = -1 / ope_nest
  )
markup_nest_summary <- reg_data %>%
  group_by(parent) %>%
  summarise(
    avg_markup_nest = mean(markup_nest, na.rm = TRUE),
    sd_markup_nest = sd(markup_nest, na.rm = TRUE),
    median_markup_nest = median(markup_nest, na.rm = TRUE),
    max_markup_nest = max(markup_nest, na.rm = TRUE),
    .groups = "drop"
```

```
) %>%
  arrange(desc(avg_markup_nest))
markup_nest_summary
```

```
# A tibble: 4 x 5
```

|   | parent<br><chr> | avg_markup_nest<br><dbl> | sd_markup_nest<br><dbl> | median_markup_nest<br><dbl> | max_markup_nest<br><dbl> |
|---|-----------------|--------------------------|-------------------------|-----------------------------|--------------------------|
| 1 | RALCORP HOL~    | 0.0930                   | 0.0259                  | 0.0881                      | 0.370                    |
| 2 | PEPSICO INC     | 0.0929                   | 0.0206                  | 0.0907                      | 0.261                    |
| 3 | KELLOGGS        | 0.0873                   | 0.0210                  | 0.0839                      | 0.210                    |
| 4 | GENERAL MIL~    | 0.0731                   | 0.0146                  | 0.0723                      | 0.311                    |

```
ggplot(markup_nest_summary, aes(x = reorder(parent, -avg_markup_nest),
                                y = avg_markup_nest)) +
  geom_bar(stat = "identity", fill = "orange", alpha = 0.7) +
  labs(title = "Average Markup by Manufacturer (Nested)",
       x = "Manufacturer", y = "Average Markup (Nested)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
ggsave("q2-9-1.pdf", width = 10, height = 6, dpi = 300)
```

## 2.10

```
# Cross - market diversion ratios between cheerios and corn flakes/froot loops
brands <- c("GENERAL MILLS CHEERIOS", "KELLOGGS CORN FLAKES", "KELLOGGS FROOT LOOPS")

row_stats <- reg_data %>%
  filter(brand %in% brands) %>%
  mutate(sj = pmin(sj, 0.999999)) %>%
  group_by(brand) %>%
  summarise(row_value = mean(-own_price_elasticity_2sls_nest * (sj / (1 - sj)), na.rm = TRUE),
            .groups = "drop")

diversion_matrix <- matrix(NA_real_, 3, 3, dimnames = list(brands, brands))
row_vals <- setNames(row_stats$row_value, row_stats$brand)

for (b in brands) {
  if (!is.na(row_vals[b])) diversion_matrix[b, setdiff(brands, b)] <- row_vals[b]
}

diversion_matrix
```

|                        | GENERAL MILLS CHEERIOS | KELLOGGS CORN FLAKES |
|------------------------|------------------------|----------------------|
| GENERAL MILLS CHEERIOS | NA                     | 0.23218833           |
| KELLOGGS CORN FLAKES   | 0.10098464             | NA                   |
| KELLOGGS FROOT LOOPS   | 0.08138226             | 0.08138226           |

|                        | KELLOGGS FROOT LOOPS |
|------------------------|----------------------|
| GENERAL MILLS CHEERIOS | 0.2321883            |
| KELLOGGS CORN FLAKES   | 0.1009846            |
| KELLOGGS FROOT LOOPS   | NA                   |

## 2.11

```
#| label: q2-11 # Find average markup for a hypothetical merger between General Mills and Kelloggs
merged_markup <- reg_data %>% mutate( parent_merged = if_else(parent %in% c("GENERAL MILLS", "KELLOGGS"), "GM_K", parent) ) %>% group_by(parent_merged) %>% summarise( avg_markup_merged = mean(markup_nest, na.rm = TRUE), .groups = "drop" ) %>% arrange(desc(avg_markup_merged)) merged_markup
```

**The average markup for the merged entity “GM\_K” is:**

```
merged_markup %>% filter(parent_merged == "GM_K") %>% pull(avg_markup_merged)
# This is higher than the individual average markups of General Mills and Kelloggs, #
indicating that the merger could lead to increased market power and higher markups df“
```